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Sentiment Analysis on Hardware product reviews

Agenda



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Pre-processing



Sentiment Analysis

Comparing lexicons

- VADER
- TextBlob

Our model

- Processed Data vs
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TF-IDF



Conclusion/insights

About the Data

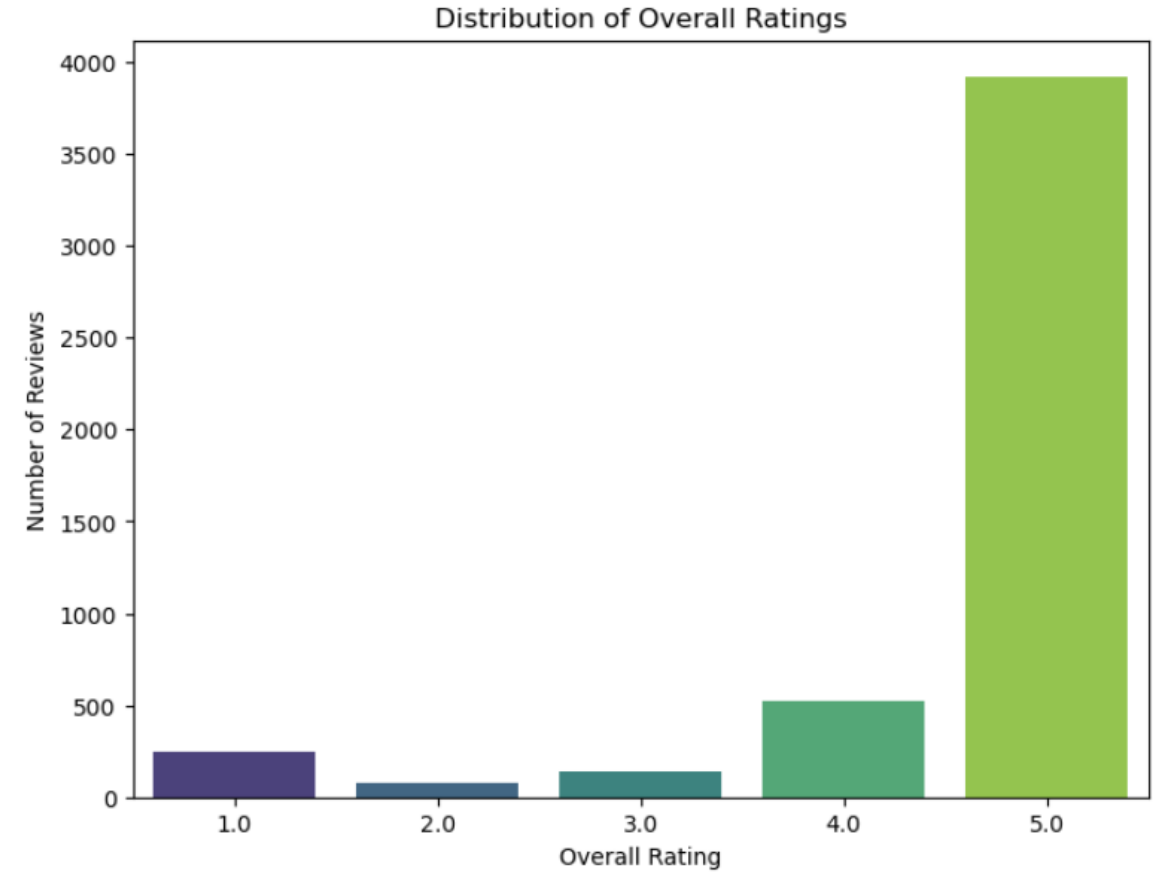
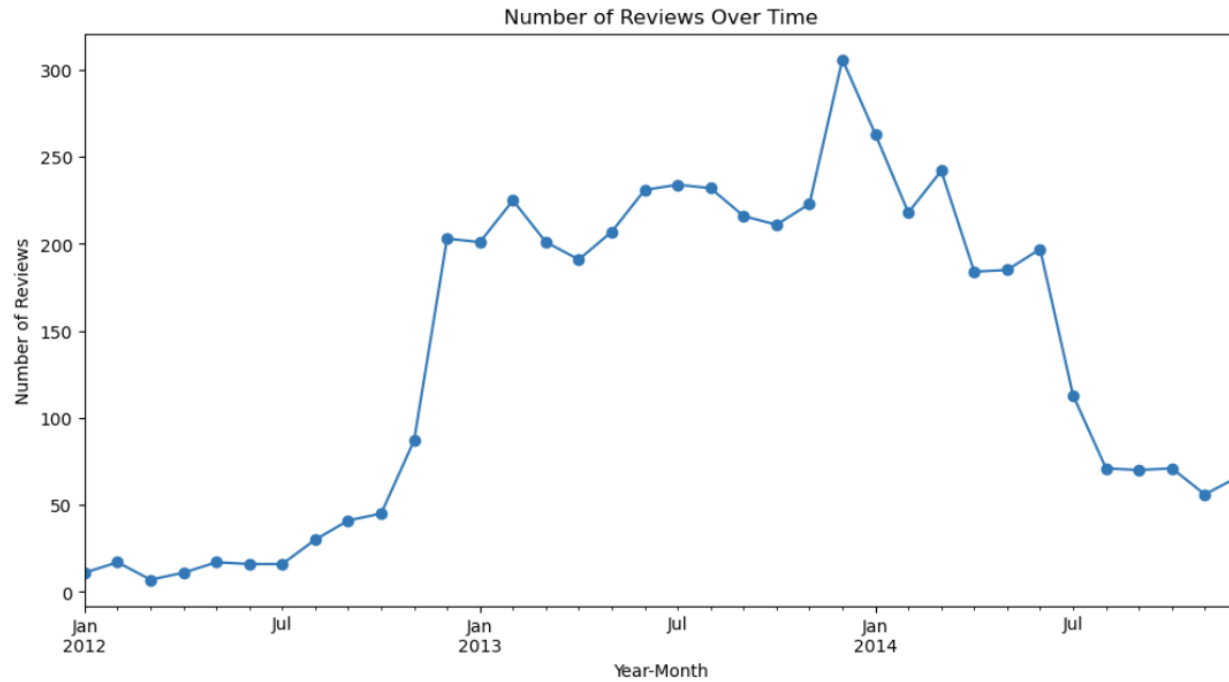
- Kaggle dataset which contains 4,914 Amazon product reviews for a specific SanDisk hard drive

Key Variables (12 total):

- ReviewerName (str) - Username that wrote review
- Overall Rating (float) - the total stars reviewer left to rate product
- ReviewText (str) - Text comment that user wrote
- ReviewTime (date) - Time at which the review was posted



About the Data



Pre-Processing

1. Cleaning- Removed special characters and numbers
2. Tokenization – Sentences were already separated. Word tokenization was done, giving us a list of lists of strings
3. POS tagging
4. Stopwords removal
5. Lemmatization



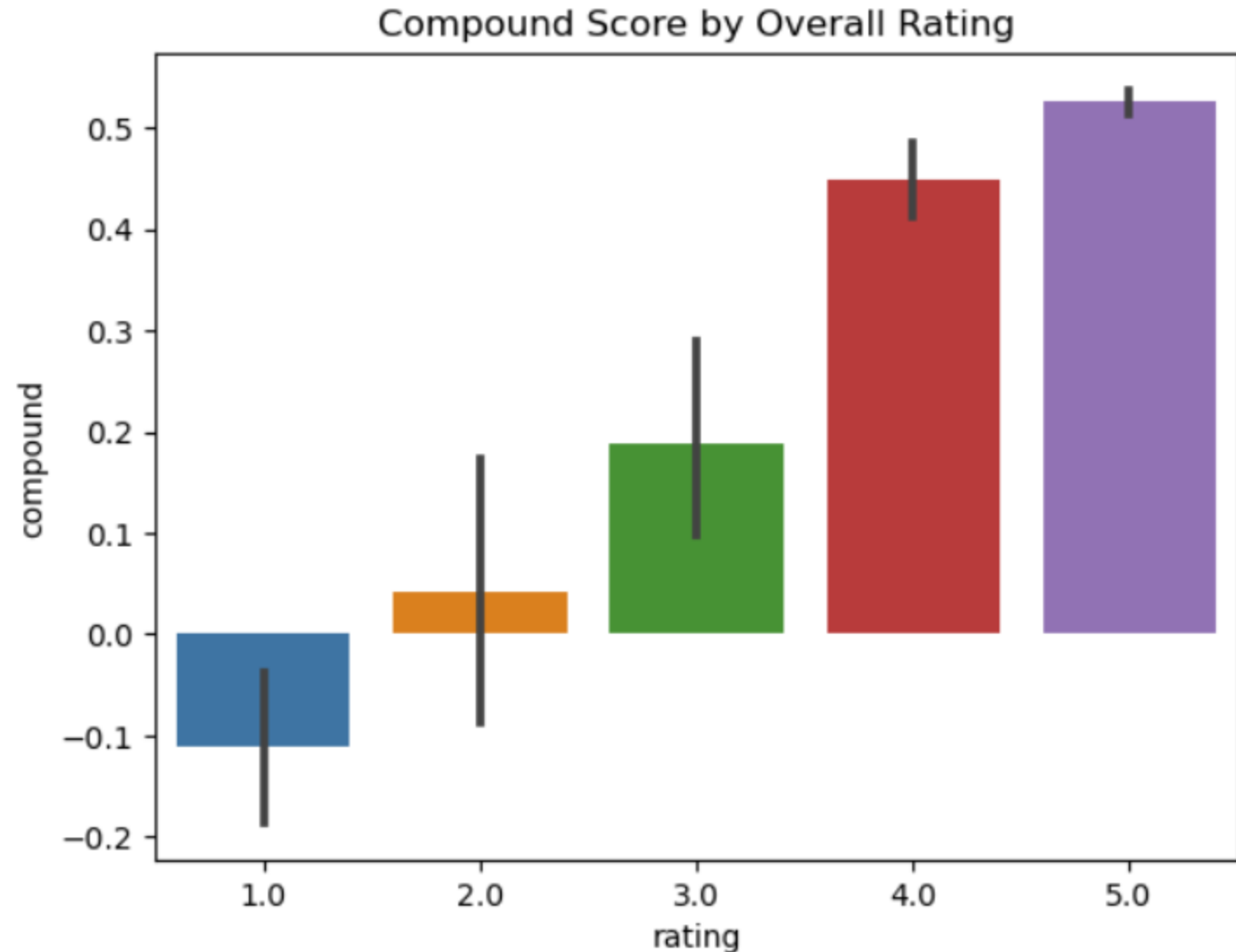
Sentiment Analysis - VADER

- VADER (Valence Aware Dictionary and Sentiment Reasoner) is a lexicon and rule-based sentiment analysis tool . It specializes in determining sentiments expressed in social media, taking into account context and punctuation. Its lexicon returns positive, negative, and neutral scores, with a reflecting compound score.
- Compound scores to comment sentiment:
 - Greater than .5 = Positive Review
 - Between 0 and .5 = Neutral Review
 - Less than 0 = Negative Review

	id	neg	neu	pos	compound	date	Originalreview	reviews	rating	vader analysis
0	1	0.0	1.000	0.000	0.0000	2014-07-23	123 No issues.	issue	4.0	Neutral
1	2	0.0	1.000	0.000	0.0000	2013-10-25	Purchased this for my device, it worked as adv...	Purchased device work advertised never much ...	5.0	Neutral
2	3	0.0	0.828	0.172	0.4019	2012-12-23	it works as expected. I should have sprung for...	work expect sprung high capacity think make ...	4.0	Neutral
3	4	0.0	0.620	0.380	0.9201	2013-11-21	This think has worked out great.Had a diff. br...	think work great diff bran gb card go south ...	5.0	Positive
4	5	0.0	0.759	0.241	0.8591	2013-07-13	Bought it with Retail Packaging, arrived legit...	Bought Retail Packaging arrive legit orange ...	5.0	Positive

Sentiment Analysis - Vader

- Visualizing compound scores against user rating
- Upward trend as expected
- The higher the compound score, the higher the user rating
- Some overlapping between 3-5 star ratings
- Most reviews are being classified as either positive or neutral.



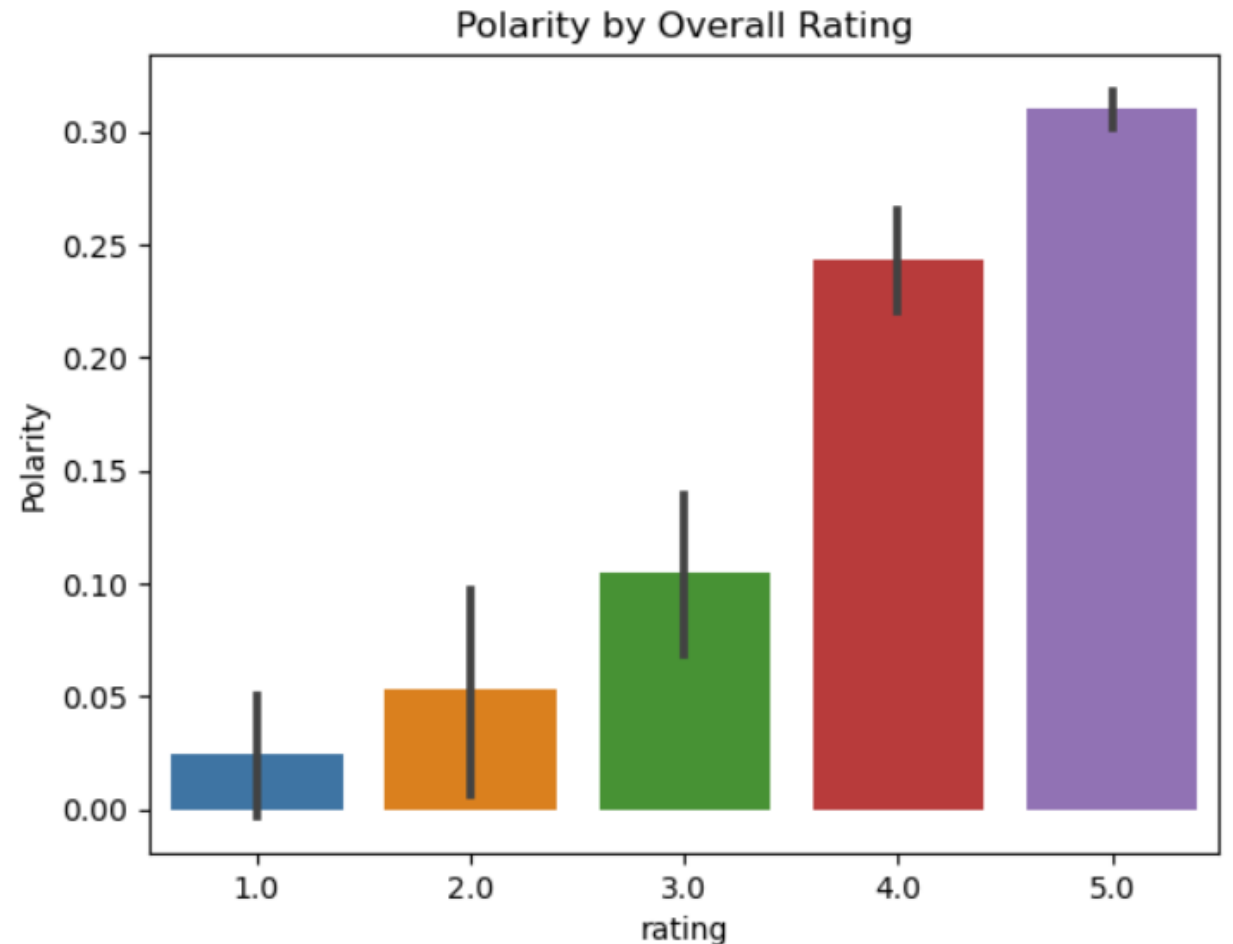
Sentiment Analysis - TextBlob

- TextBlob analyzes the sentiment in texts, focusing closely on the words and their POS tags. Its lexicon returns subjectivity and polarity scores.
- Polarity scores map to comment sentiment:
 - Greater than 0= Positive Review
 - Equals to 0 = Neutral Review
 - Less than 0 = Negative Review

	id	date	Originalreview	reviews	rating	Subjectivity	Polarity	Analysis
0	1	2014-07-23	123 No issues.	issue	4.0	0.000000	0.000000	Neutral
1	2	2013-10-25	Purchased this for my device, it worked as adv...	Purchased device work advertised never much ...	5.0	0.200000	-0.100000	Negative
2	3	2012-12-23	it works as expected. I should have sprung for...	work expect sprung high capacity think make ...	4.0	0.635000	0.031667	Positive
3	4	2013-11-21	This think has worked out great.Had a diff. br...	think work great diff bran gb card go south ...	5.0	0.616667	0.350000	Positive
4	5	2013-07-13	Bought it with Retail Packaging, arrived legit...	Bought Retail Packaging arrive legit orange ...	5.0	0.360000	0.386667	Positive

Sentiment Analysis - Vader

- Visualizing polarity scores against user rating
- Also has an upward trend
- There is a lot of overlap. This shows the model had a hard time distinguishing the sentiment of the reviews.
- A large majority of the reviews classified as positive



Comparing Lexicon results - Vader vs. TextBlob

- There's a big difference when comparing the value counts between the VADER analysis and TextBlob analysis.
- Just looking at the first three reviews, both lexicons agree the first one is neutral, however, VADER returns the next two as neutral as well while TextBlob returns them as positive.
- "works as advertised" and "works as expected" to me is more of a neutral statement so I would say VADER is more accurate in those cases.
- Comparing the bar plots proves VADER does a better job at distinguishing as well.

```
In [31]: VADER['vader analysis'].value_counts()
```

```
Out[31]: vader analysis
Positive    2940
Neutral     1735
Negative     240
Name: count, dtype: int64
```

```
In [40]: BLOB.Analysis.value_counts()
```

```
Out[40]: Analysis
Positive    4132
Negative     451
Neutral     332
Name: count, dtype: int64
```

Naive Bayes – Training and Test data creation

- Used a 70/30 split of our data to create a training and test data set
- Training dataset contains 3440 reviews
- Test dataset contains 1475 reviews

```
In [75]: x_train, x_test, y_train, y_test = train_test_split(dfClean["reviews"], dfClean["rating"], test_size=0.3, random_state=42)
```

```
In [76]: print("Shape of training data (x_train):", x_train.shape)
print("Shape of testing data (x_test):", x_test.shape)
print("Shape of training labels (y_train):", y_train.shape)
print("Shape of testing labels (y_test):", y_test.shape)
```

```
Shape of training data (x_train): (3440,)
Shape of testing data (x_test): (1475,)
Shape of training labels (y_train): (3440,)
Shape of testing labels (y_test): (1475,)
```

Naive Bayes – Processed reviews

Accuracy: 0.9335593220338984

Classification Report:

	precision	recall	f1-score	support
-1	0.74	0.56	0.64	105
0	0.00	0.00	0.00	40
1	0.94	0.99	0.97	1330
accuracy			0.93	1475
macro avg	0.56	0.52	0.54	1475
weighted avg	0.90	0.93	0.92	1475

- 93.35% accuracy
- Low support for negative reviews and neutral reviews
- Model has a high precision classifying positive reviews but is not as precise with negative reviews
- Model has trouble classifying neutral reviews

Naive Bayes – Unprocessed Reviews

- 93.22% accuracy – slightly lower than previous model
- Also has low support for negative and neutral reviews
- Precision for positive reviews remained the same, however, precision for negative reviews went up
- This model also had trouble classifying neutral reviews

Accuracy: 0.9322033898305084

Classification Report:

	precision	recall	f1-score	support
-1	0.80	0.52	0.63	105
0	0.00	0.00	0.00	40
1	0.94	0.99	0.97	1330
accuracy			0.93	1475
macro avg	0.58	0.51	0.53	1475
weighted avg	0.90	0.93	0.92	1475

TF-IDF

- Utilized NLP techniques with spaCy and NLTK.
- TF-IDF (Term Frequency-Inverse Document Frequency) used for feature extraction.
- Measures the importance of terms in documents relative to the entire corpus.
- Utilized scikit-learn's TfidfVectorizer for TF-IDF vectorization.
- Split the dataset into training and testing sets (80-20 split).
- Transformed the raw text data into numerical vectors.

Naive Bayes Classification

Accuracy: 0.90

Classification Report:

	precision	recall	f1-score	support
-1	0.00	0.00	0.00	70
0	0.00	0.00	0.00	25
1	0.90	1.00	0.95	888
accuracy			0.90	983
macro avg	0.30	0.33	0.32	983
weighted avg	0.82	0.90	0.86	983



Tested the classifier on the TF-IDF vectorized testing data.



Calculated accuracy, precision, recall, and F1-score for model evaluation.



Results:



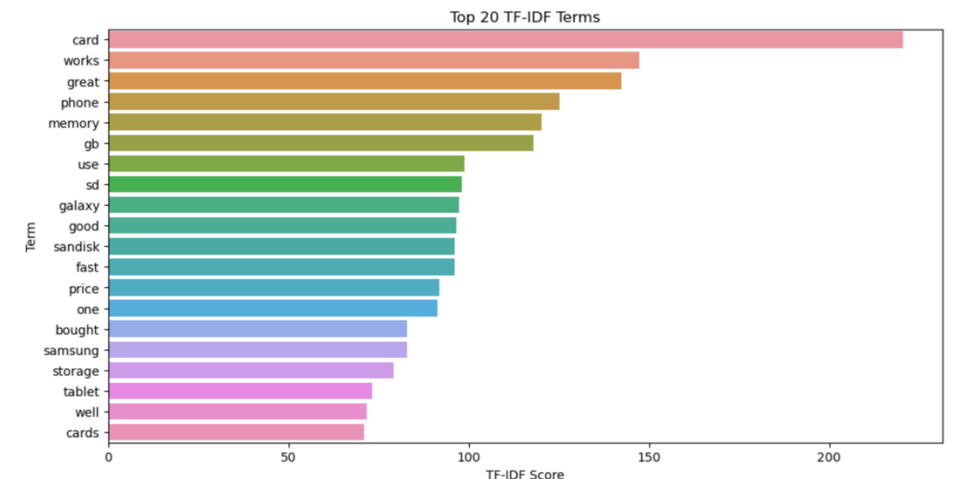
Achieved an accuracy of 90%, indicating the model's overall correctness.



The model demonstrates high precision, recall, and F1-score for positive sentiment (1).



Limited success in predicting negative sentiment (-1), likely due to imbalanced data.



Insights and Takeaways

- Might want to use a larger dataset in the future + dataset with more even representation of negative and positive reviews
- Not enough neutral reviews to properly train the model to make that classification
- Slightly higher model accuracy with processed reviews

